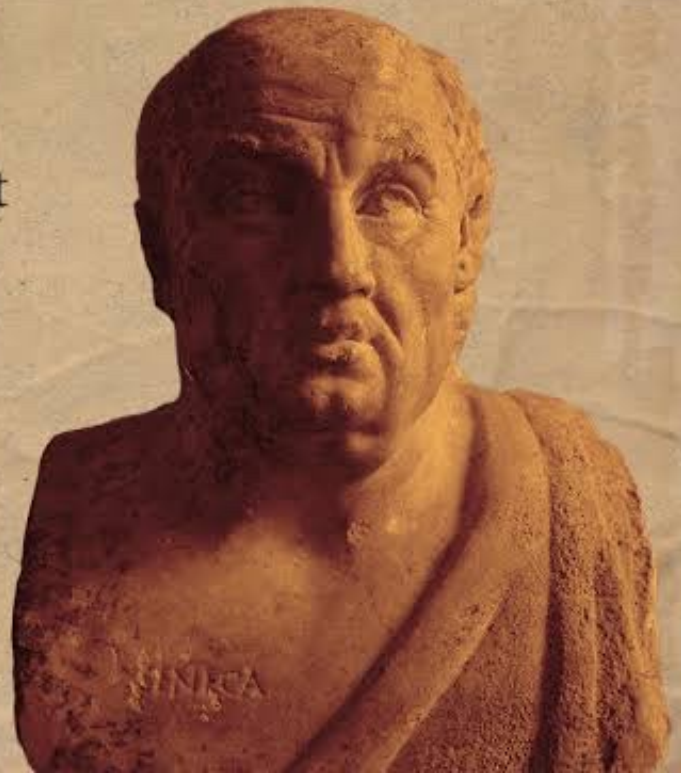


“If a man knows not to which port
he sails, no wind is favorable.”

- Seneca

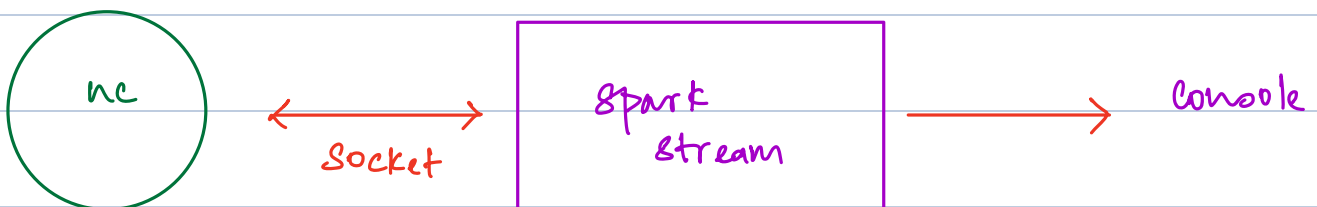
StoicQuotes.com



Agenda

- ① Spark - Structured Streaming [output types]
 - ② checkpointing in Structured Streaming [Array, Window, DateTime Fu]
-

- ① Introduction to Kafka
- ② Kafka Architecture
- ③ Components of Kafka
- ④ Demo of Kafka

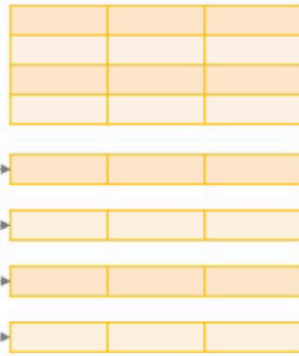


Structured Streaming Output types

Data stream



Unbounded Table



new data in the
data stream
=
new rows appended
to a unbounded table

Data stream as an unbounded table

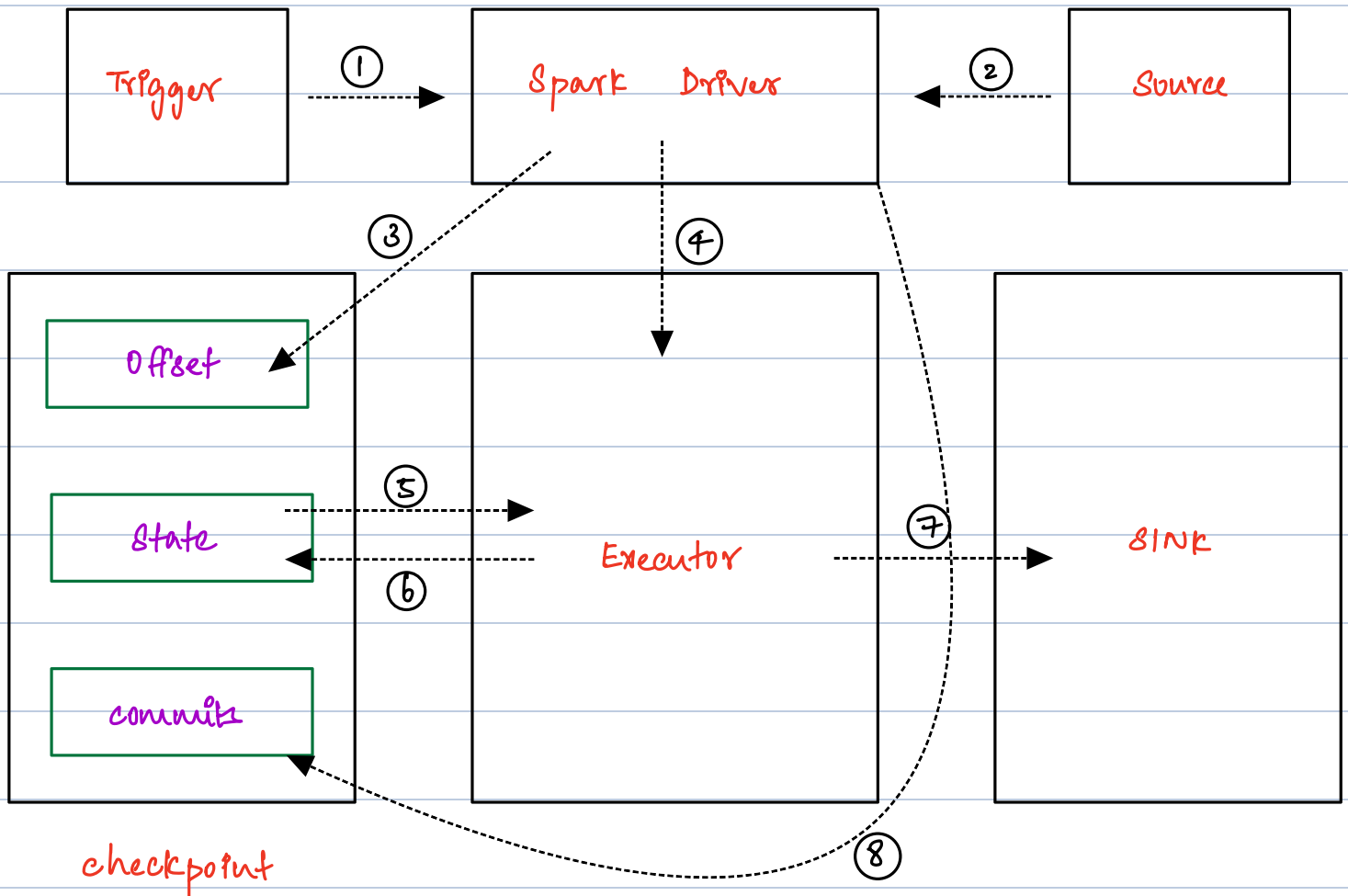
Output

- ① complete
- ② Append
- ③ update

Message	complete	Append	update
hi hi hi how are you	hi:3 you:1 how:1 are:1	hi:3 you:1 how:1 are:1	hi:3 you:1 how:1 are:1
hi how are you	hi:4 you:2 how:2 are:2	hi:1 you:1 how:1 are:1	hi:4 you:2 how:2 are:2
hi	hi:5 you:2 how:2 are:2	hi:1	hi:5

checkpoints in spark

Fault tolerance & State Management



checkpoint is usually a HDFS path

Introduction to Kafka



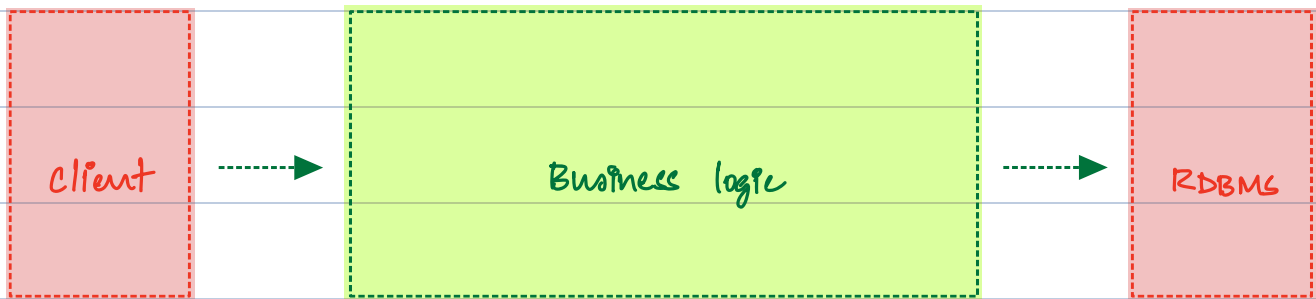
How many companies are listed in NSE & BSE?

5500 companies

How many orders are fulfilled by NSE and BSE everyday?

4,00,00,00,000 + orders

Throughput for MySQL and RDBMS → 100,000 per second

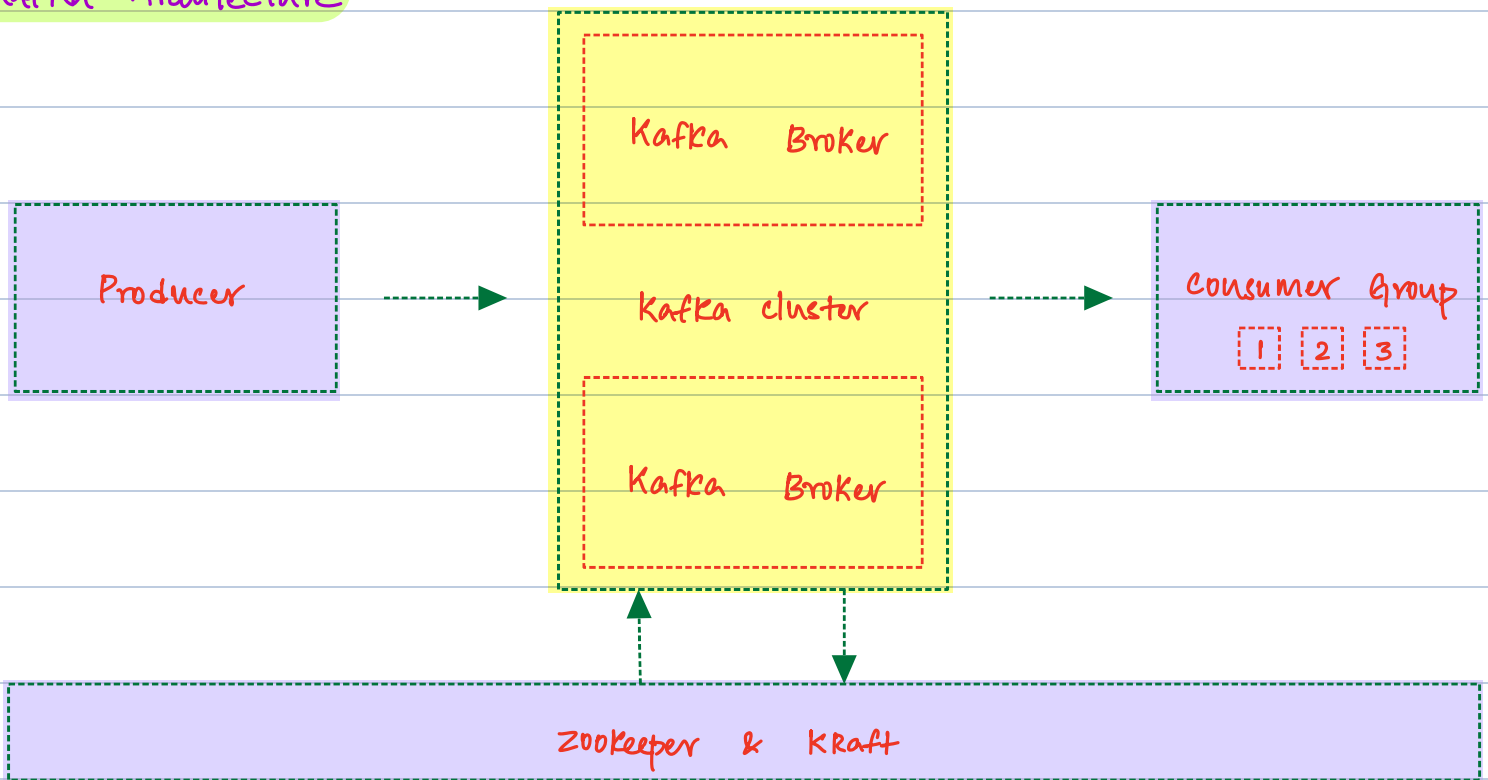


Kafka provides high throughput

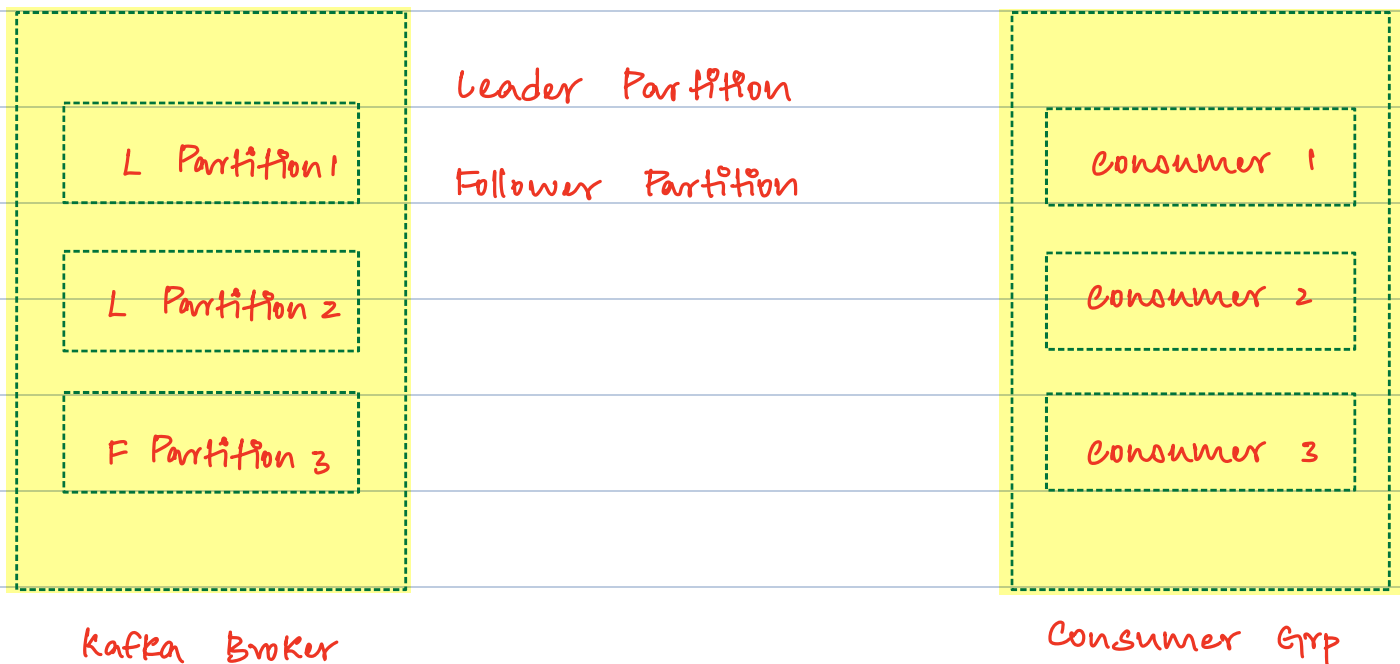
Kafka is an event streaming engine, linked in the 2011

9:59 pm → 10:06 pm

Kafka Architecture



Zooming In



AWS - Kinesis

GCP - pub/sub

If things can go wrong, they will go wrong

Components in Kafka [messaging queue, storage layer, streaming engine]

Producer - Data stream source

Broker - Kafka server

Cluster - multiple Kafka servers that are connected

Topic - category that separates and identifies data

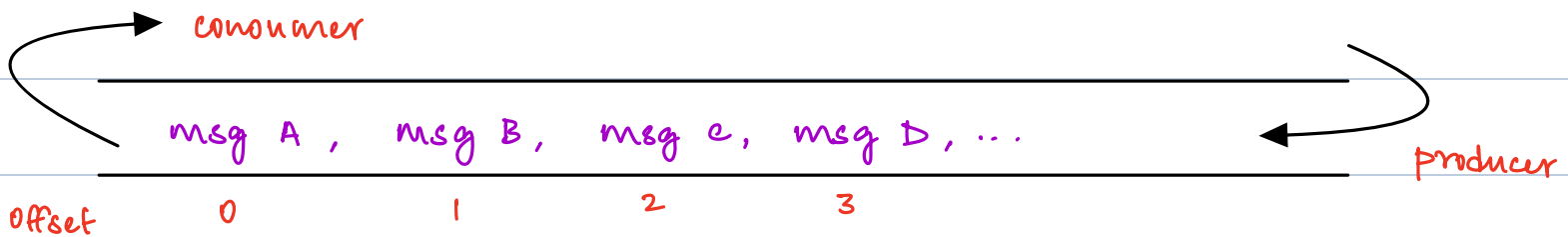
Partition - logical unit of data that are manageable in your Broker

Offset -

Consumer - Your stream destination

zookeeper - Services that continuously monitors your Kafka cluster

Internal working of Kafka



Realtime processing, Batch processing, New Monitoring service

offset → 3

1

0

Demo of Kafka

① Start my Kafka service

② Create the topic

③ Start my producer → python

④ Start my consumer